

HOMEWORK 7
ELEMENTARY LINEAR ALGEBRA
(MATH 2250-03), SPRING, 2018

CHECK: 04/12/2019

SECTION 4.5 - THE DIMENSION OF A VECTOR SPACES

1. Let H be a nonzero subspace of a finite-dimensional vector space V and let $T : V \rightarrow W$ be a linear transformation. Show that $T(H) = \{T(h) : h \in H\}$ is a subspace of W and prove that $\dim(T(H)) \leq \dim(H)$.

To show that $T(H)$ is a subspace of W , we need only show that:

- (1) $0 \in T(H)$, (2) $T(H)$ is closed under addition, (3) $T(H)$ is closed under scalar multiplication.

To show (1), note that since T is a linear map, $T(0) = 0$. So, $0 \in T(H)$. To show (2), note that if $u, v \in H$, then $u + v \in H$ (as H is a subspace) and $T(u + v) \in T(H)$. So, since $T(u) + T(v) = T(u + v)$ by the linearity of T , $T(u) + T(v) \in T(H)$ for all $T(u), T(v) \in T(H)$ and $T(H)$ is closed under addition. Finally, to show (3), note that if $T(u) \in T(H)$ and c is a scalar, then the linearity of T gives that $cT(u) = T(cu)$. Since H is a subspace, $cu \in H$ for all $u \in H$, so $T(cu) = cT(u) \in T(H)$ and $T(H)$ is closed under scalar multiplication. Therefore, we may conclude that $T(H)$ is a subspace of W .

To show that $\dim(T(H)) \leq \dim(H)$, let's recall that $\ker(T)$ is a subspace of H . So, let $\mathcal{D} = \{d_1, \dots, d_m\}$ be a basis for $\ker(T)$. Since $\ker(T)$ is a subspace of H and $\mathcal{D}$ is a basis of $\ker(T)$, it follows that $\mathcal{D}$ is a linearly independent set in H and may be expanded to a basis $\mathcal{B} = \{d_1, \dots, d_m, b_1, \dots, b_n\}$ of H (this is Theorem 11 in Section 4.5 of the textbook). So, as $\mathcal{B}$ is a basis of H , every vector $x \in H$ may be written uniquely as a linear combination of the elements of $\mathcal{B}$. That is, for every vector $x \in H$, there is a unique set of scalars $c_1, \dots, c_m, c_{m+1}, \dots, c_{m+n}$ such that

$$x = c_1 d_1 + \dots + c_m d_m + c_{m+1} b_1 + \dots + c_{m+n} b_n.$$

Since T is a linear operator and $\mathcal{D}$ is a basis for $\ker(T)$, this then says that every $T(x) \in T(H)$ satisfies

$$\begin{aligned} T(x) &= T(c_1 d_1 + \dots + c_m d_m + c_{m+1} b_1 + \dots + c_{m+n} b_n) \\ &= T(c_1 d_1 + \dots + c_m d_m) + c_{m+1} T(b_1) + \dots + c_{m+n} T(b_n) \\ &= c_{m+1} T(b_1) + \dots + c_{m+n} T(b_n). \end{aligned}$$

So, $T(H) = \text{span}\{T(b_1), \dots, T(b_n)\}$. To show that $\mathcal{E} = \{T(b_1), \dots, T(b_n)\}$ is a basis for $T(H)$, we then need only show that $\mathcal{E}$ is linearly independent. This is relatively straightforward: Since $\mathcal{B}$ is a basis of H , $\mathcal{B}$ is linearly independent and the only scalars $k_1, \dots, k_n$ which satisfy

$$k_1 b_1 + \dots + k_n b_n \in \text{span}\{d_1, \dots, d_m\} = \ker(T)$$

are $k_1 = \dots = k_n = 0$. So, $T(k_1 b_1 + \dots + k_n b_n) = 0$ if and only if $k_1 = \dots = k_n = 0$, and by the linearity of T , this says that $k_1 T(b_1) + \dots + k_n T(b_n) = 0$ if and only if $k_1 = \dots = k_n = 0$. Hence, $\mathcal{E}$ is linearly independent. So, $\mathcal{E}$ is a basis for $T(H)$, and our desired conclusion follows by noting that

$$\dim(T(H)) = \dim(\text{span}\{T(b_1), \dots, T(b_n)\}) = n \leq m + n = \dim(\text{span}\{d_1, \dots, d_m, b_1, \dots, b_n\}) = \dim(H).$$

2. Let $M_{n \times n}(\mathbb{R})$ be the vector space of all $n \times n$ matrices with real entries, and let $Sym_n(\mathbb{R})$ denote the subset of $n \times n$ matrices which are equal to their transpose, i.e.

$$Sym_n(\mathbb{R}) = \{A \in M_{n \times n}(\mathbb{R}) : A = A^T\}.$$

Such matrices are generally called *symmetric matrices*. Show that $Sym_n(\mathbb{R})$ is a vector space and compute $\dim(Sym_n(\mathbb{R}))$.

To show that $Sym_n(\mathbb{R})$ is a vector space, it is sufficient to show that $Sym_n(\mathbb{R})$ is a subspace of $M_{n \times n}(\mathbb{R})$. To do so, we need only show that:

- (1) $0 \in Sym_n(\mathbb{R})$,
- (2) $Sym_n(\mathbb{R})$ is closed under addition,
- (3) $Sym_n(\mathbb{R})$ is closed under scalar multiplication.

To show (1), let 0 denote the zero matrix and note that $0^T = 0$. So, $0 \in Sym_n(\mathbb{R})$.

To show (2), note that if $A, B \in Sym_n(\mathbb{R})$, then $(A + B)^T = A^T + B^T = A + B$. So, $A + B \in Sym_n(\mathbb{R})$ for all $A, B \in Sym_n(\mathbb{R})$, and $Sym_n(\mathbb{R})$ is closed under addition.

Finally, to show (3), note that if $A \in Sym_n(\mathbb{R})$ and c is a scalar, then $(cA)^T = cA^T = cA$. So, $cA \in Sym_n(\mathbb{R})$ for all $A \in Sym_n(\mathbb{R})$, and $Sym_n(\mathbb{R})$ is closed under scalar multiplication.

Therefore, we may conclude that $Sym_n(\mathbb{R})$ is a subspace of $M_{n \times n}(\mathbb{R})$. As a subspace of a vector space is, itself, a vector space, we have shown that $Sym_n(\mathbb{R})$ is a vector space.

To compute the dimension of $Sym_n(\mathbb{R})$, let's start by noting that if a_{ij} is the entry in the i^{th} row and j^{th} column of an $n \times n$ matrix A , then a_{ji} is the entry in the i^{th} row and j^{th} column of A^T . More succinctly: Suppose $A \in Sym_n(\mathbb{R})$. Then, since $A = A^T$, it must be the case that $a_{ij} = a_{ji}$ for all $i, j \in \{1, \dots, n\}$. For entries on the diagonal of A , this amounts to $a_{ii} = a_{ii}$, so we may choose the elements on the diagonal however we like. For elements not on the diagonal, the condition $a_{ij} = a_{ji}$ tells us that if we an entry belonging to the triangular portion of A above the diagonal, then this choice determines an entry in the triangular portion of A below the diagonal and vice versa. So, since there are n^2 entries in an $n \times n$ matrix and n entries on the diagonal, there are $n^2 - n$ entries not on the diagonal of A . As the above argument tells us that half of these remaining entries are determined by the constraint $a_{ij} = a_{ji}$, this then says that A is completely determined by the n entries on the diagonal of A and the $\frac{n^2 - n}{2}$ entries not on the diagonal of A . So, A is completely determined by exactly $\frac{n^2 - n}{2} + n = \frac{n^2 + n}{2} = \frac{n(n + 1)}{2}$ of its entries and we may conclude that

$$\dim(Sym_n(\mathbb{R})) = \frac{n(n + 1)}{2}.$$

SECTION 4.6 - RANK

3. Let $A = \begin{bmatrix} 1 & -4 & 9 & -7 \\ -1 & 2 & -4 & 1 \\ 5 & -6 & 10 & 7 \end{bmatrix}$. The matrix A is row-equivalent to the matrix $\begin{bmatrix} 1 & 0 & -1 & 5 \\ 0 & -2 & 5 & -6 \\ 0 & 0 & 0 & 0 \end{bmatrix}$.

Without calculations, compute $\text{rank}(A)$ and $\dim(\text{Nul}(A))$. Then, find bases for $\text{Col}(A)$, $\text{Row}(A)$, and $\text{Nul}(A)$.

Recall that the rank of a matrix is the dimension of its column space, and recall that the pivot columns of a matrix form a basis for its column space. So, since A is row-equivalent to the echelon form matrix

$$\begin{bmatrix} 1 & 0 & -1 & 5 \\ 0 & -2 & 5 & -6 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

and since this matrix has two pivots, $\text{rank}(A) = \dim(\text{Col}(A)) = 2$. Using the Rank Theorem, since A is a 3×4 matrix, $\text{rank}(A) + \dim(\text{Nul}(A)) = 4$. So, we may see that $\dim(\text{Nul}(A)) = 4 - 2 = 2$.

Now, let's find bases for $\text{Col}(A)$, $\text{Row}(A)$, and $\text{Nul}(A)$:

To find a basis for the column space of A, recall that the pivot columns of a matrix form a basis for its column space. So, using the row-equivalent matrix given above, $\left\{ \begin{bmatrix} 1 \\ -1 \\ 5 \end{bmatrix}, \begin{bmatrix} -4 \\ 2 \\ -6 \end{bmatrix} \right\}$ is a basis for $\text{Col}(A)$.

To find a basis for the row space of A, recall that if B is an echelon form which is row-equivalent to A, then the nonzero rows of B form a basis for the row space of A. So, using the row-equivalent

matrix given above, $\left\{ \begin{bmatrix} 1 \\ 0 \\ -1 \\ 5 \end{bmatrix}, \begin{bmatrix} 0 \\ -2 \\ 6 \\ -6 \end{bmatrix} \right\}$ is a basis for $\text{Row}(A)$.

To find a basis for $\text{Nul}(A)$, we simply solve $Ax = 0$. Since solutions to linear systems are preserved under elementary row operations, we may use the row-equivalent matrix given above to do this more easily by solving the equivalent system

$$\begin{aligned} x_1 - x_3 + 5x_4 &= 0 \\ -2x_2 + 5x_3 - 6x_4 &= 0 \end{aligned}$$

In parametric vector form, this has the solution

$$\begin{bmatrix} x_3 - 5x_4 \\ 5/2x_3 - 3x_4 \\ x_3 \\ x_4 \end{bmatrix} = x_3 \begin{bmatrix} 1 \\ 5/2 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -5 \\ -3 \\ 0 \\ 1 \end{bmatrix}.$$

So, $\left\{ \begin{bmatrix} 1 \\ 5/2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -5 \\ -3 \\ 0 \\ 1 \end{bmatrix} \right\}$ is a basis for $\text{Nul}(A)$.

4. Suppose A is an $m \times n$ matrix with real entries and suppose $b \in \mathbb{R}^m$. What must be true of $\text{rank}\left(\begin{bmatrix} A & b \end{bmatrix}\right)$ and $\text{rank}(A)$ in order for the equation $Ax = b$ to be consistent?

Let $a_1, \dots, a_n \in \mathbb{R}^m$ be the columns of A so that $A = \begin{bmatrix} a_1 & \dots & a_n \end{bmatrix}$. Then, $Ax = b$ has a solution (that is, $Ax = b$ is consistent) if and only if there exists a vector $x \in \mathbb{R}^n$ such that

$$Ax = \begin{bmatrix} a_1 & \dots & a_n \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = x_1 a_1 + \dots + x_n a_n = b.$$

That is, the equation $Ax = b$ has a solution if and only if b is a linear combination of the columns of A . Said another way, if $Ax = b$ has a solution, then $b \in \text{Col}(A)$. But, since $\text{Col}(A) = \text{span}\{a_1, \dots, a_n\}$, then it must be the case that $\text{span}\{a_1, \dots, a_n, b\} = \text{span}\{a_1, \dots, a_n\}$ whenever $Ax = b$ has a solution. Therefore, noting that

$$\text{rank}(A) = \dim(\text{Col}(A)) = \dim(\text{span}\{a_1, \dots, a_n\})$$

and

$$\text{rank}\left(\begin{bmatrix} A & b \end{bmatrix}\right) = \dim(\text{Col}\left(\begin{bmatrix} A & b \end{bmatrix}\right)) = \dim(\text{span}\{a_1, \dots, a_n, b\}),$$

we may conclude that if $Ax = b$ is consistent, then

$$\text{rank}(A) = \text{rank}\left(\begin{bmatrix} A & b \end{bmatrix}\right).$$

SECTION 4.7 - CHANGE OF BASIS

5. In parts (a) and (b) of this exercise, you may assume that $\mathcal{B} = \{b_1, b_2\}$ and $\mathcal{D} = \{d_1, d_2\}$ are bases for $\mathbb{R}^2$. Find the change-of-coordinates matrix from $\mathcal{B}$ to $\mathcal{D}$.

(a) $b_1 = \begin{bmatrix} 7 \\ 5 \end{bmatrix}, b_2 = \begin{bmatrix} -3 \\ -1 \end{bmatrix}, d_1 = \begin{bmatrix} 1 \\ -5 \end{bmatrix}, d_2 = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$

Part (a)

To find the change-of-coordinates matrix from $\mathcal{B}$ to $\mathcal{D}$, we simply form the augmented matrix $\left[\begin{array}{cc|cc} d_1 & d_2 & b_1 & b_2 \end{array} \right]$ and row-reduce:

$$\left[\begin{array}{cc|cc} d_1 & d_2 & b_1 & b_2 \end{array} \right] = \left[\begin{array}{cc|cc} 1 & -2 & 7 & -3 \\ -5 & 2 & 5 & -1 \end{array} \right] \xrightarrow{\text{Row Reduce}} \left[\begin{array}{cc|cc} 1 & 0 & -3 & 1 \\ 0 & 1 & -5 & 2 \end{array} \right].$$

So, the change-of-coordinates matrix from $\mathcal{B}$ to $\mathcal{D}$ is given by $\begin{bmatrix} -3 & 1 \\ -5 & 2 \end{bmatrix}$.

To find the change-of-coordinates matrix from $\mathcal{D}$ to $\mathcal{B}$, we could invert the change-of-coordinates matrix found above to produce

$$\begin{bmatrix} -3 & 1 \\ -5 & 2 \end{bmatrix}^{-1} = \begin{bmatrix} -2 & 1 \\ -5 & 3 \end{bmatrix}.$$

Alternatively, we could simply form the augmented matrix $\left[\begin{array}{cc|cc} b_1 & b_2 & d_1 & d_2 \end{array} \right]$ and row-reduce:

$$\left[\begin{array}{cc|cc} b_1 & b_2 & d_1 & d_2 \end{array} \right] = \left[\begin{array}{cc|cc} 7 & -3 & 1 & -2 \\ 5 & -1 & -5 & 2 \end{array} \right] \xrightarrow{\text{Row Reduce}} \left[\begin{array}{cc|cc} 1 & 0 & -2 & 1 \\ 0 & 1 & -5 & 3 \end{array} \right].$$

So, using either approach, we may see that the change-of-coordinates matrix from $\mathcal{D}$ to $\mathcal{B}$ is given by

$$\begin{bmatrix} -2 & 1 \\ -5 & 3 \end{bmatrix}.$$

$$(b) \ b_1 = \begin{bmatrix} -6 \\ -1 \end{bmatrix}, b_2 = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, d_1 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, d_2 = \begin{bmatrix} 6 \\ -2 \end{bmatrix}$$

Part (b)

To find the change-of-coordinates matrix from $\mathcal{B}$ to $\mathcal{D}$, we simply form the augmented matrix $\left[\begin{array}{cc|cc} d_1 & d_2 & b_1 & b_2 \end{array} \right]$ and row-reduce:

$$\left[\begin{array}{cc|cc} d_1 & d_2 & b_1 & b_2 \end{array} \right] = \left[\begin{array}{cc|cc} 2 & 6 & -6 & 2 \\ -1 & -2 & -1 & 0 \end{array} \right] \xrightarrow{\text{Row Reduce}} \left[\begin{array}{cc|cc} 1 & 0 & 9 & -2 \\ 0 & 1 & -4 & 1 \end{array} \right].$$

So, the change-of-coordinates matrix from $\mathcal{B}$ to $\mathcal{D}$ is given by $\begin{bmatrix} 9 & -2 \\ -4 & 1 \end{bmatrix}$.

To find the change-of-coordinates matrix from $\mathcal{D}$ to $\mathcal{B}$, we could invert the change-of-coordinates matrix found above to produce

$$\begin{bmatrix} 9 & -2 \\ -4 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 2 \\ 4 & 9 \end{bmatrix}.$$

Alternatively, we could simply form the augmented matrix $\left[\begin{array}{cc|cc} b_1 & b_2 & d_1 & d_2 \end{array} \right]$ and row-reduce:

$$\left[\begin{array}{cc|cc} b_1 & b_2 & d_1 & d_2 \end{array} \right] = \left[\begin{array}{cc|cc} -6 & 2 & 2 & 6 \\ -1 & 0 & -1 & -2 \end{array} \right] \xrightarrow{\text{Row Reduce}} \left[\begin{array}{cc|cc} 1 & 0 & 1 & 2 \\ 0 & 1 & 4 & 9 \end{array} \right].$$

So, using either approach, we may see that the change-of-coordinates matrix from $\mathcal{D}$ to $\mathcal{B}$ is given by

$$\begin{bmatrix} 1 & 2 \\ 4 & 9 \end{bmatrix}.$$

6. In $\mathbb{P}_2$, find the change-of-coordinates matrix from the basis $\mathcal{B} = \{1 - 2t + t^2, 3 - 5t + 4t^2, 2t + 3t^2\}$ to the standard basis $\mathcal{C} = \{1, t, t^2\}$. Then, using this matrix, find the $\mathcal{B}$ -coordinate vector $[p]_{\mathcal{B}}$ for the polynomial $p(t) = -1 + 2t$.

Recall that the change-of-coordinates matrix from the basis $\mathcal{B} = \{b_1, b_2, b_3\}$ to the basis $\mathcal{C}$ is given by

$$\begin{bmatrix} [b_1]_{\mathcal{C}} & [b_2]_{\mathcal{C}} & [b_3]_{\mathcal{C}} \end{bmatrix}.$$

So, since

$$[b_1]_{\mathcal{C}} = [1 - 2t + t^2]_{\mathcal{C}} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}, \quad [b_2]_{\mathcal{C}} = [3 - 5t + 4t^2]_{\mathcal{C}} = \begin{bmatrix} 3 \\ -5 \\ 4 \end{bmatrix}, \quad [b_3]_{\mathcal{C}} = [2t + 3t^2]_{\mathcal{C}} = \begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix},$$

it then follows that the change-of-coordinates matrix from $\mathcal{B}$ to $\mathcal{C}$ is given by

$$\begin{bmatrix} [b_1]_{\mathcal{C}} & [b_2]_{\mathcal{C}} & [b_3]_{\mathcal{C}} \end{bmatrix} = \begin{bmatrix} 1 & 3 & 0 \\ -2 & -5 & 2 \\ 1 & 4 & 3 \end{bmatrix}.$$

Recall that this matrix satisfies

$$[q]_{\mathcal{C}} = \begin{bmatrix} [b_1]_{\mathcal{C}} & [b_2]_{\mathcal{C}} & [b_3]_{\mathcal{C}} \end{bmatrix} [q]_{\mathcal{B}}$$

for all $q \in \mathbb{P}_2$. So, since

$$[p(t)]_{\mathcal{C}} = [-1 + 2t]_{\mathcal{C}} = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix},$$

we may find $[p(t)]_{\mathcal{B}}$ by carrying out the following computation

$$\begin{aligned} [p(t)]_{\mathcal{B}} &= \begin{bmatrix} [b_1]_{\mathcal{C}} & [b_2]_{\mathcal{C}} & [b_3]_{\mathcal{C}} \end{bmatrix}^{-1} [p(t)]_{\mathcal{C}} \\ &= \begin{bmatrix} 1 & 3 & 0 \\ -2 & -5 & 2 \\ 1 & 4 & 3 \end{bmatrix}^{-1} \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} -23 & -9 & 6 \\ 8 & 3 & -2 \\ -3 & -1 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix}. \end{aligned}$$

If you'd like to check that this is correct, it's easy to verify that:

$$(5)(1 - 2t + t^2) + (-2)(3 - 5t + 4t^2) + (1)(2t + 3t^2) = -1 + 2t.$$